

# MIT 18.06 Exam 1 (solved)

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## Question 1

(a) Possible values for  $m$ ,  $n$ , and rank  $r$  of  $A$

We have  $Ax = [1 \ 0]$  has no solutions, and  $Ax = [1 \ 1]$  has exactly one solution.

So, the right-hand side is a 2-vector, but the solution says  $m = 3$  (so the right-hand side should be 3-vectors). That means  $A$  is  $3 \times n$ .

If  $Ax = b$  has no solution for some  $b$ ,  $A$  can't be full rank (not all  $b$  are in the column space), so  $r < m$ . But  $Ax = b$  has exactly one solution for some  $b$ , so  $A$  must have  $r = n$  (so the nullspace is trivial). So  $r = n < m$ , so  $n = r = 1$  or  $2$ , and  $m = 3$ .

So possible:  $m = 3$ ,  $n = 1$  or  $2$ ,  $r = n$ .

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(b) All solutions to  $Ax = 0$

If  $r = n$ , then the nullspace is just  $x = 0$ . So the only solution is  $x = 0$ .

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(c) Example of a matrix  $A$  that fits the description

Let  $A = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$  (so  $3 \times 2$ , rank 2).

Check:  $Ax = [1 \ 0 \ 1]$ , try  $x = [1 \ 0]$ , get  $[1 \ 0 \ 1]$ , so that works. But  $Ax = [1 \ 0 \ 0]$ , no solution (since third row is sum of first two, but right side doesn't match).

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## Question 2

Given: The  $3 \times 3$  matrix  $A$  reduces to the identity matrix  $I$  by the following row operations (in order):

- $E_{21}$ : Subtract 4 times row 1 from row 2.
  - $E_{31}$ : Subtract 3 times row 1 from row 3.
  - $E_{23}$ : Subtract row 3 from row 2.
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(a) Write  $A^{-1}$  in terms of the  $E$ 's and compute  $A^{-1}$

**Solution:**

To get  $A^{-1}$ , since  $E_{23}E_{31}E_{21}A = I$ ,  $A^{-1} = E_{23}E_{31}E_{21}$ .

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 & -4 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} E_{31} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & -3 & 0 & 1 \end{bmatrix}$$

$$E_{23} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 \end{bmatrix}$$

Multiply them in order:

$$\text{First } E_{31}E_{21}: \begin{bmatrix} 1 & 0 & 0 & -4 & 1 & 0 & -3 & 0 & 1 \end{bmatrix}$$

Now  $E_{23}$  times that: Row 1: stays the same. Row 2:  $(-4, 1, 0) - (-3, 0, 1) = (-1, 1, -1)$  Row 3: stays the same.

$$\text{So } A^{-1} = \begin{bmatrix} 1 & 0 & 0 & -1 & 1 & -1 & -3 & 0 & 1 \end{bmatrix}$$

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(b) What is the original matrix  $A$ ?

**Solution:**

To get  $A$ , invert the  $E$  matrices (add instead of subtract):  $E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 4 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$

$$E_{31}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 3 & 0 & 1 \end{bmatrix} E_{23}^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Multiply in order: First  $E_{31}^{-1}E_{23}^{-1}$ : Row 1:  $(1, 0, 0)$  Row 2:  $(0, 1, 1)$  Row 3:  $(3, 0, 1)$

Now  $E_{21}^{-1}$  times that: Row 1:  $(1, 0, 0)$  Row 2:  $4 \times (1, 0, 0) + (0, 1, 1) = (4, 1, 1)$  Row 3:  $(3, 0, 1)$

$$\text{So } A = \begin{bmatrix} 1 & 0 & 0 & 4 & 1 & 1 & 3 & 0 & 1 \end{bmatrix}$$

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(c) What is the lower triangular factor  $L$  in  $A = LU$ ?

**Solution:**

$L$  is just the product of the inverses of the  $E$  matrices (the ones with the multipliers below the diagonal):

$$L = \begin{bmatrix} 1 & 0 & 0 & 4 & 1 & 0 & 3 & 0 & 1 \end{bmatrix}$$

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## Question 3

Given: The  $3 \times 4$  matrix  $A$  depends on  $c$ :

$$A = \begin{bmatrix} 1 & 1 & 2 & 4 & 3 & c & 2 & 8 & 0 & 0 & 2 & 2 \end{bmatrix}$$

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(a) For each  $c$ , find a basis for the column space of  $A$

**Solution:**

Let  $a_1 = [1 \ 3 \ 0]$ ,  $a_2 = [1 \ c \ 0]$ ,  $a_3 = [2 \ 2 \ 2]$ ,  $a_4 = [4 \ 8 \ 2]$ .

If  $c \neq 3$ ,  $a_1, a_2, a_3$  are independent (since  $a_2$  isn't a multiple of  $a_1$  unless  $c = 3$ ).  $a_4$  can be written as a combination of the others. So basis:  $a_1, a_2, a_3$ .

If  $c = 3$ ,  $a_2 = a_1$ , so basis:  $a_1, a_3$ .

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(b) For each  $c$ , find a basis for the nullspace of  $A$

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**Solution:**

Set up  $Ax = 0$ :

1.  $x_1 + x_2 + 2x_3 + 4x_4 = 0$
2.  $3x_1 + cx_2 + 2x_3 + 8x_4 = 0$
3.  $2x_3 + 2x_4 = 0$  so  $x_3 = -x_4$

Plug  $x_3 = -x_4$  into the first two:

1.  $x_1 + x_2 + 2(-x_4) + 4x_4 = x_1 + x_2 + 2x_4 = 0$
2.  $3x_1 + cx_2 + 2(-x_4) + 8x_4 = 3x_1 + cx_2 + 6x_4 = 0$

Let  $x_4 = t$  (free). From (1):  $x_1 = -x_2 - 2t$ . Plug into (2):

$$3(-x_2 - 2t) + cx_2 + 6t = -3x_2 - 6t + cx_2 + 6t = (c - 3)x_2 = 0$$

So if  $c \neq 3$ ,  $x_2 = 0$ , so  $x_1 = -2t$ ,  $x_3 = -t$ ,  $x_4 = t$ . Nullspace basis:  $(-2, 0, -1, 1)$ .

If  $c = 3$ ,  $x_2$  is free, so basis:  $(-1, 1, 0, 0)$  and  $(-2, 0, -1, 1)$ .

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(c) For each  $c$ , find the complete solution  $x$  to  $Ax = [1 \ c \ 0]$

Let  $x_4 = t$  (free).  $x_3 = -t$ . From the first equation:  $x_1 + x_2 + 2(-t) + 4t = x_1 + x_2 + 2t = 1$ , so  $x_1 = 1 - x_2 - 2t$ . Second equation:  $3x_1 + cx_2 + 6t = c$ . Plug  $x_1$  in:  $3(1 - x_2 - 2t) + cx_2 + 6t = c$   
 $3 - 3x_2 - 6t + cx_2 + 6t = c$   $3 + (c - 3)x_2 = c$  So  $(c - 3)x_2 = c - 3$

If  $c \neq 3$ ,  $x_2 = 1$ ,  $x_1 = 1 - 1 - 2t = -2t$ ,  $x_3 = -t$ ,  $x_4 = t$ . Take  $t = 0$  for a particular solution:  $(0, 1, 0, 0)$ . General solution:  $(0, 1, 0, 0) + \alpha(-2, 0, -1, 1)$ .

If  $c = 3$ ,  $x_2$  is free,  $x_1 = 1 - x_2 - 2t$ ,  $x_3 = -t$ ,  $x_4 = t$ . Take  $t = 0$ :  $(1 - x_2, x_2, 0, 0)$ . General solution:  $(1 - x_2, x_2, 0, 0) + \beta(-1, 1, 0, 0) + \gamma(-2, 0, -1, 1)$ .

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## Question 4

(a) If  $A$  is a  $3 \times 5$  matrix, what information do you have about the nullspace of  $A$ ?

**Solution:**

$A$  is  $3 \times 5$ , so  $Ax = 0$  for  $x \in \mathbb{R}^5$ . The nullspace has dimension  $5 - r$ , and  $r \leq 3$ , so nullspace is at least 2-dimensional.

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(b) Suppose row operations on  $A$  lead to  $R = \text{rref}(A)$ :

$$R = \begin{bmatrix} 1 & 4 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Write all known information about the columns of  $A$ .

**Solution:**

Looking at the rref, pivots are in columns 1, 4, 5. So columns 1, 4, 5 of  $A$  are linearly independent and form a basis for the column space. Column 2 is 4 times column 1, and column 3 is zero.

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(c) In the vector space  $M$  of all  $3 \times 3$  matrices, what subspace  $S$  is spanned by all possible row reduced echelon forms  $R$ ?

**Solution:**

All possible RREFs for  $3 \times 3$  matrices are upper triangular, so the subspace  $S$  is the set of all  $3 \times 3$  upper triangular matrices (dimension 6).